

A Practical Core Loss Calculation Method of Filter Inductors in PWM Inverters Based on the Modified Steinmetz Equation

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Abstract—In this paper, a practical core loss calculation method based on the modified Steinmetz equation is proposed. It calculates the low-frequency losses and the high-frequency losses separately. By utilizing the curves of permeability versus DC bias provided by core vendors instead of plenty tests, the paper establishes a relation between the DC bias and the core losses. A novel model of minor loops is proposed to calculate the high-frequency losses based on the simulation results and Steinmetz parameters of cores. The low-frequency loss is analyzed in depth from the hysteresis curves in the B - H plane. The calculated results are verified by measuring the core losses of the filter inductor in a three-level NPC inverter. The proposed model can be derived without any test. Meanwhile, the detailed low-frequency loss calculation is presented. Moreover, the influence of DC bias is also taken into consideration to improve the accuracy. The proposed method is practical and accurate to predict the core losses.

Keywords—core loss; DC bias; minor loop; Steinmetz equation; PWM inverter.

I. INTRODUCTION

For designing power electronic systems, predicting the core losses is essential. The Steinmetz equation is widely used as:

$$P = kf^\alpha B^\beta \quad (1)$$

where P is the time-average power loss per unit volume, B is the peak flux density amplitude, f is the frequency of a sinusoidal excitation, and k , α , β are material parameters found by curve fitting.

The core losses of sinusoidal voltage excitations can be well evaluated in limited flux densities and frequencies. However, DC bias and the shape of voltage waveforms can affect the core losses significantly [1]-[4] in power electronics applications especially for the PWM converters.

To overcome this drawback and determine the core losses with non-sinusoidal excitations, different kinds of hysteresis models have been developed. The Preisach model, strongly related to the mechanisms of magnetic hysteresis, is improved in [5]. The Jiles-Atherton hysteresis model is proposed based on macroscopic energy calculation in [6]. There are other kinds of equivalent resistor models proposed in [7-8]. Although these models can characterize the hysteresis phenomenon, the complexity makes them impractical in calculating the core losses in PWM inverters.

In recent decades, some practical approaches are proposed in [9]-[11] based on the modified Steinmetz equation: the MSE is analyzed in [9] according to the thought that core loss is dependent on dB/dt ; the GSE and the iGSE are established and improved in [10] and [11]. The iGSE is not only proved more accurate to calculate the core losses of arbitrary waveforms but also practical for the circuit design since the parameters can be obtained from the data sheet of cores and circuit simulations. However, these methods neglect the fact that core losses varies in different DC bias conditions and may not be effective for calculating the core losses of the filter inductor in PWM inverters.

In addition, a practical hysteresis loss calculation method with PWM voltage excitations is proposed and improved in [12]-[16]. It separates core losses into two parts: the high-frequency losses and the low-frequency losses. A loss map is created by measuring the minor loops on the B - H plane to calculate the high-frequency losses so that the influence of DC bias and PWM voltage waveforms can be obtained. Thus, by summing up minor loops losses of a filter inductor in PWM inverters, the high-frequency losses can be obtained. Owing to accurate measurement of the minor loops of the inductor cores, the calculation method is simple and precise which results in its widely adopted in PWM circuits.

However, an additional “buck” or other circuit is needed to obtain the loss map to predict the high-frequency losses; the low-frequency core loss calculation method is not well clarified which is significant to predict the core losses.

This paper develops a practical and accurate model to predict core losses without any test. With this approach, the high-frequency losses and the low-frequency losses can be distinguished clearly by minor loops and major loops respectively. The influence of DC bias is taken into account with the curves of permeability versus DC bias rather than plenty experiment, and the influence of triangle current waveforms on core losses is also discussed. The low-frequency losses are analyzed in detail. For verifying the proposed models, an accurate and convenient measurement setup of core losses is built. A three-level NPC inverter setup was built for testing the core losses. In order to improve the interference immunity to the electric and magnetic fields, a coaxial resistor and a coaxial cable are applied in this platform. The proposed method is easy and precise to predict core losses only with the data sheet of cores and circuit simulations.

II. PROPOSED METHOD TO MEASURE THE CORE LOSSES

Many core loss measurement methods have been discussed in [17]. The B - H loop measurement method is evaluated as a suitable approach since it is accurate and fast. A secondary winding is necessary to test voltages due to the copper losses are not included in the measurement as shown in Fig. 1. To obtain the B - H loop, the voltage and the current of the filter inductor should be acquired. Fig. 1 depicts a core loss test platform of a three-level NPC inverter. According to the measuring principle of the B - H loop method, the current through the test resistor and the voltage of the secondary winding should be tested. Typical current and voltage waveforms are illustrated in Fig. 2. The principle of calculating the core losses can be expressed by the following equations:

$$H(t) = \frac{N_1 \times i_p(t)}{l_e} \quad (2)$$

$$B(t) = \frac{1}{A_e \times N_2} \int_0^t u_s(\tau) d\tau \quad (3)$$

$$P = \int_0^T \frac{N_1}{N_2} u_s i_p dt = A_e l_e \int_{B(0)}^{B(T)} H dB \quad (4)$$

Where, N_1 is the turns of the excitation windings in the inductor, N_2 is the turns of the voltage sense winding, A_e , l_e are the effective cross sectional area and the effective path length of magnetic core respectively, i_p is the current of the primary windings and v_s is the voltage of the secondary windings. It is indicated by Equation (4) that the core losses in a period are proportional to the area of B - H curve.

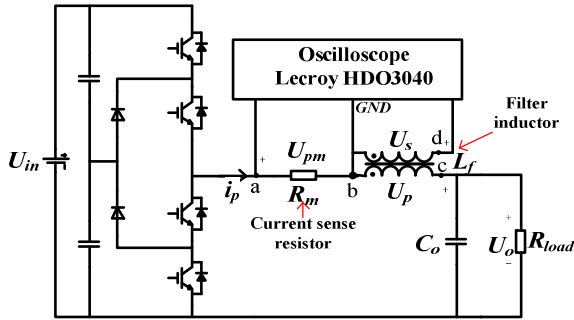


Fig. 1. A three-level NPC inverter test system (“GND” is the common ground of the oscilloscope)

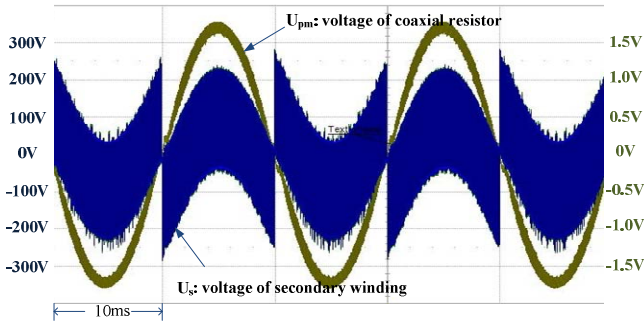


Fig. 2. The current and voltage waveforms of the filter inductor (the switching frequency is 15kHz)

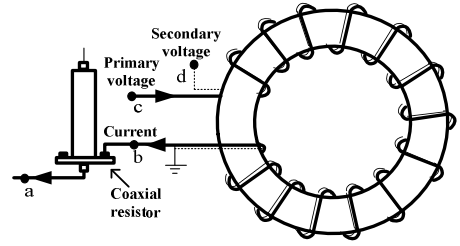


Fig. 3. The coaxial resistor and the core under test

There are several ways for the current measurement as discussed in [18-19]. The current probe is typically used to measure current signals ranging from DC to high frequencies on account of high bandwidth. However, it is quite sensitive to the electromagnetic environment. The magnetic field around the filter inductor will bring much interference. Other than the current probe, the current sense resistor is also utilized to detect core losses. Due to its fast frequency response, the current sense resistor performs well in measuring core losses. A coaxial resistor in Fig. 3 with the value of 0.03964Ω is employed because of the energy consumption considerations.

The advantages of the measurement equipment of the core losses are:

- By utilizing a strand of the primary windings as the secondary winding rather than an extra winding, the best coupling can be obtained.
- The probes are common voltage probes with extremely high bandwidth (500MHz).
- The interference of electromagnetic environment can be limited as small as possible by using the coaxial resistor and the coaxial cable.
- Using the common connection point “GND” as the oscilloscope common ground as shown in Fig. 1, the voltage probes can be the same type. Another reason of employing the common ground is that some experiment shows the deviations of test core losses are very large by using the differential probes.

Table I summarizes the inductor parameters. Using the equations (2) and (3), the B - H plane can be displayed on the oscilloscope as shown in Fig. 4.

TABLE I. INDUCTOR SPECIFICATIONS

Core material	Fe-Ni powder core (High flux)
Core shape	Toroid
Initial inductance	$L=800\mu\text{H}$
Excitation turn number	$N_1=42$
Sense turn number	$N_2=42$
Effective cross-sectional area	$A_e=1416\text{ mm}^2$
Effective path length	$l_e=200\text{ mm}$
Windows area	$W_e=1808\text{ mm}^2$

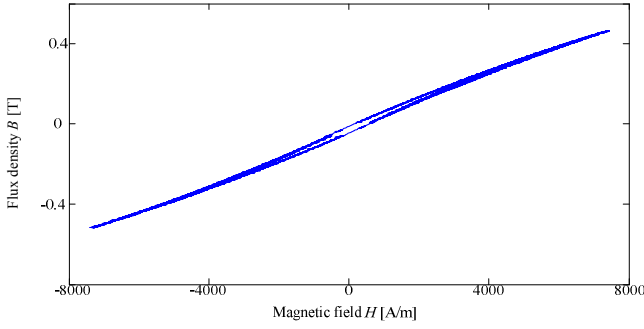


Fig. 4. B-H curve of the core

III. PROPOSED CORE LOSSES MODELS

The proposed core losses calculation method divides the total core losses into the high-frequency losses caused by the high-frequency ripples and the low-frequency losses owing to the low-frequency current. The high-frequency losses models are established by the modified Steinmetz equation. The influence of DC bias on the high-frequency losses is analyzed by utilizing the curves of the relative incremental permeability versus DC bias provided by core vendors. The influence of the triangle current waveforms on the high-frequency losses is modeled by the iGSE method. The low-frequency loss is analyzed from the hysteresis curves in the B - H plane and modeled by the modified Steinmetz equation.

A. High-Frequency Losses Model

Before calculating the high-frequency losses, the influence of DC bias on core losses should be tested. By applying different DC magnetic field H_{dc} and AC flux density B_{ac} to a powder core, the losses are illustrated as shown in Fig. 5. The experimental principle is similar to Fig. 2 in [1]. From the test results, the core losses increase slowly when DC bias rises from 0 to 5000 A/m, but dramatically when it is beyond 5000 A/m.

Fig. 6 depicts the normalized increasing ratios of the core losses. The reciprocal of relatively permeability with different DC bias offered by manufacturers are plotted with H_{dc} as well. The rising trends of all the curves with different B_{ac} are similar to each other as shown in Fig. 6. As H_{dc} increases, the working condition of cores approaches the saturation regions where core losses rise and the relative permeability drops. For the industrial applications, the curves of relative incremental permeability versus DC bias can be employed to correct the influence of DC bias on core losses. The curve-fit formulas provided by core vendors as the denominator of equation (5) are usually polynomial. By adopting equation (5), the core losses with DC bias can be calculated by equation (6).

$$f_{dc}(H_{dc}) = \frac{\mu_0}{\mu} = \frac{1}{aH_{dc}^4 + bH_{dc}^3 + cH_{dc}^2 + dH_{dc} + e} \quad (0 < H_{dc} < 10000 \text{ A/m}) \quad (5)$$

$$P_{dc} = f_{dc}(H_{dc}) \times P_{ac} = f_{dc}(H_{dc}) \times k \times f_s^\alpha \times B_{ac}^\beta \quad (6)$$

where, f_{dc} is the factor function of DC bias, μ_0 is the permeability without DC bias, μ is the permeability with different DC bias, a, b, c, d, e are the coefficients which are provided in data sheet, P_{dc} is the loss with different DC bias, P_{ac} is the loss without DC bias, k, α, β are the Steinmetz parameters offered by data sheet, B_{ac} is the peak flux density amplitude and f_s is the frequency of sinusoidal excitations. The loss of different DC bias is fitted as the functions of B_{ac} and H_{dc} .

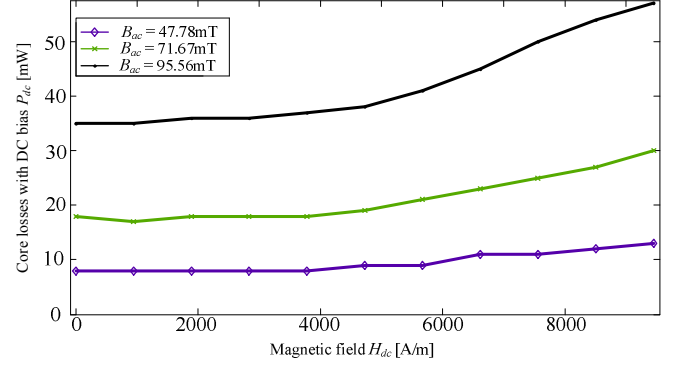


Fig. 5. Core losses under DC bias condition, $f = 10\text{kHz}$

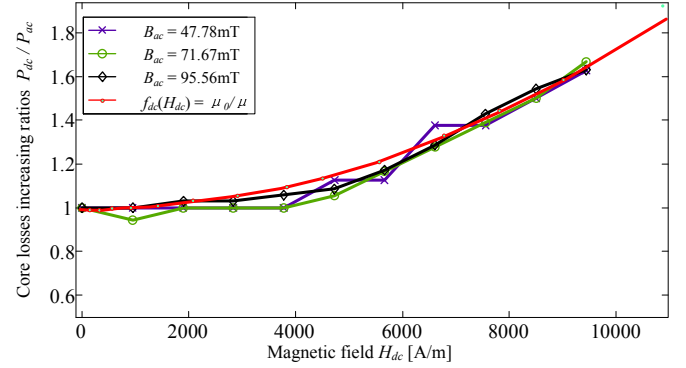


Fig. 6. Increasing ratios of the losses and the reciprocal of permeability change rate under different DC bias

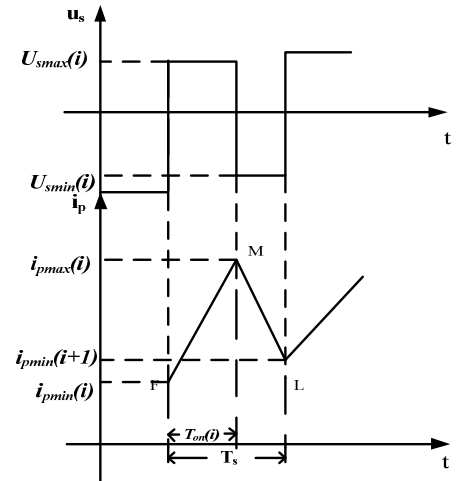


Fig. 7. Simulation waveforms of voltage(up) and current(down) in AC filter inductor in one switching period

The shape of voltage waveforms can also influence core losses. Simulation waveforms of the voltage and the current in the filter inductor in one switching period are shown in Fig. 7. The current waveforms are triangle with different rise time $T_{on}(i)$, which can be characterized with the duty of the voltage in equation (7). The iGSE method proposed in [11] is accurate to calculate core losses of arbitrary waveforms without DC bias. According to this method, the equations of calculating the core losses of triangle current waveforms can be developed as presented in equation (8).

$$D(i) = \frac{T_{on}(i)}{T_s} \quad (7)$$

$$f_{tr}(i) = \frac{P_{tr}(i)}{P_{sin}(i)} = \frac{D(i)^\alpha + [1 - D(i)]^{1-\alpha}}{(2\pi)^{\alpha-1} \int_0^{2\pi} |\cos\theta|^\alpha \cdot 2^{\beta-\alpha} d\theta} \quad (8)$$

where, f_{tr} is the factor function, i means the i th switching period, T_s is the switching period, k , α , β are the Steinmetz parameters offered by core data sheet, P_{tr} is the core loss of triangle current waveforms and P_{sin} is the sinusoidal core loss in one switching period.

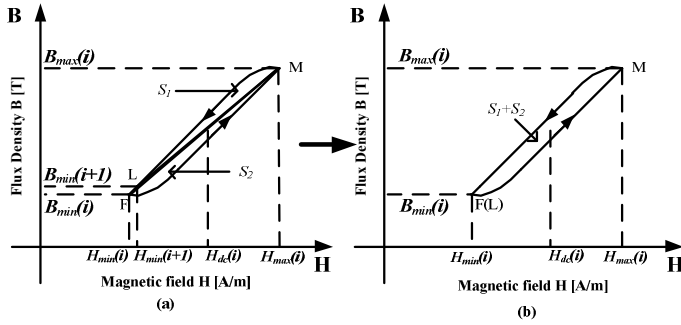


Fig. 8. An approximate rectangular hysteresis minor loop model. (a) Open-loop trajectory. (b) Equivalent close-loop trajectory.

The influence of DC bias and triangle current waveforms on the core losses with a closed-loop trajectory is taken into consideration by combining equation (6) with equation (8). Fig. 8 (a) shows the minor loop of one switching period. Because the current values of $i_{pmin}(i)$ and $i_{pmin}(i+1)$ in Fig. 7 are different, the loop in Fig. 8(a) is open. The high-frequency loss of this loop is proportional to the sum areas of S_1 and S_2 . Since point F is near to point L , the sum areas can be regarded as a closed loop as shown in Fig. 8 (b). So the total high-frequency losses can be calculated by the equations (9)-(12):

$$n = \frac{f_s}{f_o} \quad (9)$$

$$B_{ac}(i) = \int (u - V_o) dt \approx (U_{smax}(i) - U_o) \cdot \Delta t = (U_{smax} - U_o) \cdot D(i) T \quad (10)$$

$$H_{dc}(i) = \frac{1}{2} \times [H_{min}(i) + H_{max}(i)] \quad (11)$$

$$P_h = \sum_{i=1}^n f_{dc}[H_{dc}(i)] \times f_{tr}(i) \times k \times f_s^{\alpha-1} \times [B_{ac}(i)]^\beta \quad (12)$$

where, f_o is the output current frequency, n is the number of switching cycles in an output cycle which can be calculated by equation (9), U_{smax} is the max voltage in inductor as shown in Fig., U_o is the output voltage, P_h is the total high frequency in one output cycle, f_s is the switching frequency in Hz.

B. Low-Frequency Losses Model

The low-frequency losses are proportional to the major loop which is surrounded by two envelopes as illustrated in Fig. 9. One envelope is composed of the points (H_{min}, B_{min}) and the points (H_{max}, B_{max}) as illustrated in Fig. 8. By using equation (10), the envelopes of B_{max} and B_{min} in time domain in an output cycle can be obtained as shown in Fig. 10.

It is worth noticing that both trajectories of (H_{min}, B_{min}) and (H_{max}, B_{max}) are under the excitations of sinusoidal waves as shown in Fig. 10, so the low-frequency losses proportional to the major loop area can be regarded as excited by a sinusoidal wave. Therefore, the modified Steinmetz equation can be applied to calculate the low-frequency losses. In the Steinmetz equation, the peak flux density should be acquired precisely. This can be expressed as the maximum flux density in the envelope of (H_{min}, B_{min}) in Fig. 10.

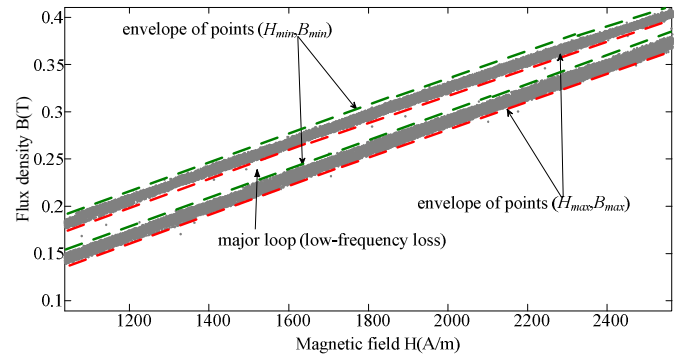


Fig. 9. The partial zoomed graph of the B-H curve

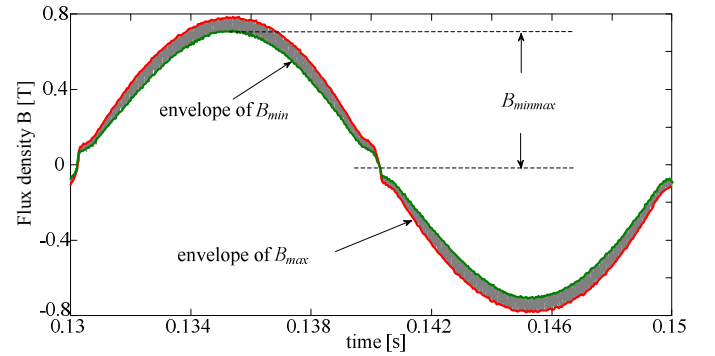


Fig. 10. The flux density waveforms in time domain

The peak density can be calculated by the equation (13), and the low-frequency losses are calculated by equation (14).

$$B_{mac} = B_{minmax} \quad (13)$$

$$P_l = k \times f_o^{\alpha-1} \times B_{mac}^\beta \quad (14)$$

where, B_{mac} is the peak flux density amplitude of major loop, f_o is the output frequency, B_{minmax} is the maximum of the envelop of (H_{min}, B_{min}) , P_l is the low-frequency loss, and other parameters is provided by core data sheet. Since all these parameters can be received from simulations or the core manufacturers, it is a convenient method.

C. Total Core Losses Calculation

The total core losses can be calculated by summing up the high-frequency losses and the low-frequency losses. With the equations (12) and (14), the total losses can be calculated by the following equations:

$$m = f_o \quad (15)$$

$$P_{tot} = \sum_{j=0}^m [P_h(j) + P_l(j)] \quad (16)$$

where, P_{tot} is the total loss in [W], j is the j th output cycle in a second, and m is the total number of output cycle in a second and is equal to output frequency f_o .

IV. EXPERIMENTAL VERIFICATION

In order to verify the high-frequency and low-frequency losses models presented in Section III, a three-level NPC inverter is built and tested accordingly as shown in Fig. 1. The specifications set for the prototype are shown in Table II. The inductor employed in this circuit is made by Fe-Ni powder (high flux) material which is suitable for high power PWM inverters. To reduce the copper losses, litz wires are used as windings in this inductor. The coaxial cable and the coaxial resistor are utilized to test the inductor current. By integrating the current and the secondary winding voltage, the total core losses can be acquired according to the equation (4). A series of experiment at different inputs are carried out to verify the loss model.

TABLE II. THE THREE-LEVEL NPC INVERTER SPECIFICATIONS

Input voltage V_i	550V
Output voltage V_o	185V
Output current I_o	20A-35A
Switching frequency f	10kHz, 15kHz, 20kHz
Oscilloscope model	LeCroy HDO3040
Voltage probe for resistor	The coaxial(0.03964Ω)
Voltage probe for secondary windings	Lecory PP018

The comparison of the test total losses of the experiment at different frequencies (10kHz, 15kHz, 20kHz) and the calculating total losses using the proposed models are illustrated in Fig. 11 (a)-(c).

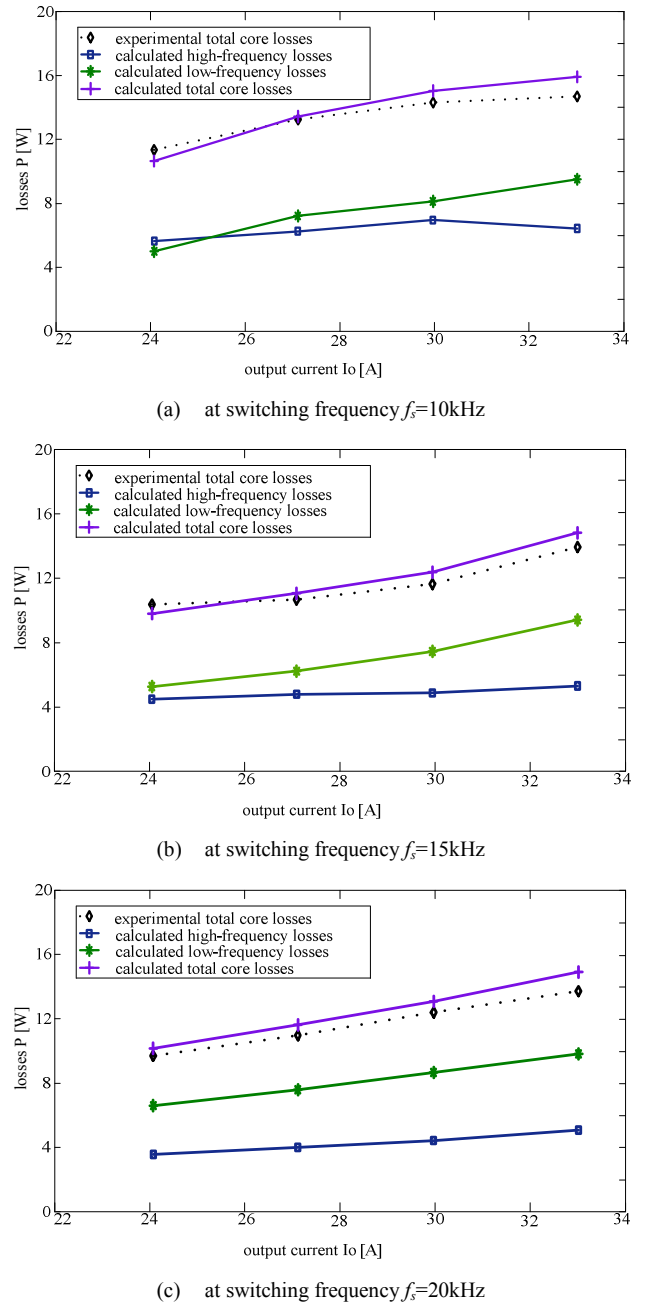


Fig. 11. The calculated high-frequency losses and the comparison of the calculated total losses and the experimental total losses at different switching frequencies (10kHz, 15kHz, 20kHz).

TABLE III. DEVIATIONS OF CALCULATED RESULTS

Output current I_o (A)	Deviations (%) $f_s=10\text{kHz}$	Deviations (%) $f_s=15\text{kHz}$	Deviations (%) $f_s=20\text{kHz}$
24	6.198	5.441	4.710
27	1.321	3.513	5.563
30	5.332	6.320	5.807
33	8.560	6.254	8.712

As shown in Fig. 11 (a)-(c), the total core losses, the high-frequency losses and the low-frequency losses become larger with the output current increasing. The largest deviations of the calculated and the experimental losses are about 8.5% (at 10kHz), 6.2% (at 15kHz) and 8.7% (at 20kHz) as shown in Table III. For circuit designs, the deviations are relatively accurate. All deviations are less than 10%, which may be caused by the following reasons:

- Deviations from loss models: Steinmetz parameters are not exactly precise by fitting curves. DC bias influence to loss is complicated in different situations. This model by utilizing the curves of permeability versus DC bias is relatively accurate. In terms of the low-frequency model, the equation (14) may be lack of adequate theoretical basis, but is quite accurate according to the experiment results.
- Errors from core loss measurement: since the electromagnetic environment can bring some interference to the probe, this error can't be eliminated totally. But the coaxial resistor is the best instrument which can minimize this interference.

The low-frequency losses account a large proportion of the total losses as can be seen from Fig. 11 (a)-(c). With the output current increasing, the high-frequency losses rise because the amplitude of the ripple currents becomes bigger. At the same time, the larger amplitude of output current leads to more low-frequency losses.

V. CONCLUSION

In this paper, a novel core loss calculation method based on the modified Steinmetz equation is proposed. This method separates the total core losses into the high-frequency losses and the low-frequency losses. With the curves of permeability versus DC bias instead of a large number of tests, the influence of DC bias on core losses can be predicted easily. Moreover, the low-frequency losses are analyzed and modeled exhaustively from the hysteresis curves in the B - H plane. By utilizing the coaxial resistor and the coaxial cable, the measurement results of core losses are very precise. The experimental results are given to demonstrate the validation of the proposed approach. All results show the method is quite efficient and practical way to predict core losses particularly applicable for the PWM inverters. Furthermore, all parameters can be achieved only from the data sheet of cores and simulations.

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